Yingru (Richard) Li

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Science of RL training at frontier scale: optimization, scaling laws, stability and efficiency.

Impact

- Core contributor to RL fundamentals at xAI for the Grok RL big runs—the science of RL training (optimization, scaling laws, stability)—**contributing to one of the longest stable RL runs to date**.
- Diagnosed LLM-RL training collapse (Sep 2025) as an off-policy bias from the training–inference mismatch; the fix—masked importance sampling (MIS), made principled as Trust Region Masking (TRM)—is **adopted, shipped, or discussed by DeepSeek-V3.2, veRL, Qwen, vLLM, SGLang, Slime, Miles, and Cognition (SWE-grep)**; widely discussed on X (thread; an **84K-view** quote-post).
- Authored the Rollout Correction framework in **veRL (20k+ GitHub stars)**, which corrects the gap between the rollout policy and the training policy (training–inference mismatch, stale asynchronous rollouts, replayed data); it is **the base on which veRL's newer off-policy algorithms (e.g., score centering and IcePop) are built**, and MIS and TRM ship there as first-class algorithms. Also authored the Optimal Token Baseline, a variance-reducing token-level baseline.

Experience

xAI (now SpaceX Super Intelligence) — Singapore
Member of Technical Staff — Dec 2025 – Present

- Core contributor to RL fundamentals for large-scale ("big-run") reinforcement learning of the Grok models.
  - *Science of RL training* — core algorithms and recipes; RL scaling from first principles; optimizer and numerics at scale.
  - *Stability and efficiency* — identified and resolved critical training instabilities affecting every stage of the RL pipeline, contributing to one of the longest stable RL runs to date.
  - *Long horizon* — training recipes for long-horizon agentic RL.
  - *On-policy distillation* — training recipes and supporting infrastructure.
  - *Infrastructure* — built and scaled these recipes in large-scale distributed RL systems.

ByteDance — Singapore
LLM Algorithm Research Scientist — 2025 – Dec 2025

- Led RL training stability and efficiency for LLM reasoning and agents.
  - Traced RL training collapse to implicit off-policyness from the training–inference mismatch; developed Trust Region Masking (ICML'26), Optimal Token Baseline (ICML'26), Dynamic Vocabulary Pruning, and a simple LR-scheduling fix.
  - Core contributor to veRL: authored the Rollout Correction framework and the Optimal Token Baseline estimator.
  - Co-led end-to-end RL for multi-turn tool-integrated reasoning (SimpleTIR, ICLR'26) and a Triton kernel-generation agent (Dr. Kernel, ICML'26).

Tencent AI & Robotics X — Shenzhen, China
Student Research Staff Member, Agent Center — 2019 – 2022

- Led work on scalable exploration in deep RL—HyperAgent (ICML'24), HyperDQN (ICLR'22)—and stable off-policy learning—Divergence-Augmented Policy Optimization (NeurIPS'19).

Microsoft Research — Beijing, China
Research Intern, Theory Center — 2016

- Combinatorial optimization and reinforcement / bandit learning algorithms.

Selected Publications

Full list on Google Scholar. *equal contribution; †corresponding author.

1. **The Optimal Token Baseline: Variance Reduction for Long-Horizon LLM-RL**
   Yingru Li, Jiawei Xu, Ziniu Li, Jiacai Liu, Wei Liu, Yuxuan Tong, Longtao Zheng, Zhenghai Xue, Yaxiang Zhang, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *ICML 2026*
2. **Trust Region Masking for Long-Horizon LLM Reinforcement Learning**
   Yingru Li, Jiacai Liu, Jiawei Xu, Yuxuan Tong, Ziniu Li, Qian Liu, Baoxiang Wang. *ICML 2026*

3. **When Speed Kills Stability: Demystifying RL Collapse from the Training-Inference Mismatch**
   Jiacai Liu*, **Yingru Li***,†, Yuqian Fu, Jiawei Wang, Qian Liu, Yu Shen. *Technical Blog, 2025*
4. **Beyond Precision: Training-Inference Mismatch is an Optimization Problem and Simple LR Scheduling Fixes It**
   Yaxiang Zhang*, **Yingru Li***,†, Jiacai Liu, Jiawei Xu, Ziniu Li, Qian Liu, Haoyuan Li. *arXiv 2026*
5. **Logit Dynamics in Softmax Policy Gradient Methods**
   Yingru Li. *arXiv 2025*
6. **Dynamic Vocabulary Pruning: Stable LLM-RL by Taming the Tail**
   Yingru Li, Jiawei Xu, Jiacai Liu, Yuxuan Tong, Ziniu Li, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *arXiv 2025*
7. **Dr. Kernel: Reinforcement Learning Done Right for Triton Kernel Generations**
   Wei Liu, Jiawei Xu, **Yingru Li**, Longtao Zheng, Tianjian Li, Qian Liu, Junxian He. *ICML 2026*
8. **SimpleTIR: End-to-End Reinforcement Learning for Multi-Turn Tool-Integrated Reasoning**
   Zhenghai Xue, Longtao Zheng, Qian Liu, **Yingru Li**, Xiaosen Zheng, Zejun Ma, Bo An. *ICLR 2026*
9. **Optimistic Thompson Sampling for No-Regret Learning in Unknown Games**
   Yingru Li, Liangqi Liu, Wenqiang Pu, Hao Liang, Zhi-Quan Luo. *IEEE Transactions on Signal Processing, 2026*
10. **Scalable Exploration via Ensemble++**
    Yingru Li, Jiawei Xu, Baoxiang Wang, Zhi-Quan Luo. *NeurIPS 2025*
11. **Q-Star Meets Scalable Posterior Sampling: Bridging Theory and Practice via HyperAgent**
    Yingru Li, Jiawei Xu, Lei Han, Zhi-Quan Luo. *ICML 2024*
12. **HyperDQN: A Randomized Exploration Method for Deep Reinforcement Learning**
    Ziniu Li, **Yingru Li**†, Yushun Zhang, Tong Zhang, Zhi-Quan Luo. *ICLR 2022*
13. **Divergence-Augmented Policy Optimization**
    Qing Wang*, **Yingru Li***, Jiechao Xiong, Tong Zhang. *NeurIPS 2019*

Selected Writing

Theory of RL training: why post-training updates are sparse and low-rank, and how asynchronous RL should scale. Posts at richardli.xyz/post.

- *The Shape of a Post-Training Update* (four parts, 2026); *Off-Policy RL at Scale* (two parts, 2026); *Information Bandwidth in Reinforcement Learning* (2025).

Education

The Chinese University of Hong Kong (CUHK) 2018 – 2025
Ph.D. in Computer and Information Engineering. Advisor: Zhi-Quan (Tom) Luo. Committee: Xinyun Chen, Baoxiang Wang, John C.S. Lui, Benjamin Van Roy (Stanford & DeepMind).

Huazhong University of Science and Technology (HUST) 2020
B.E./M.E. in Computer Science. Ranked 1st/134 overall; 1st/26 in Software and Theoretical CS.

Selected Honors

- ICML 2026 Gold Reviewer; reviewer for NeurIPS, ICML, ICLR, AISTATS, and TMLR (40+ papers).
- Best Student Paper, IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM), 2024.
- SRIBD Ph.D. Fellowship (Gold Class), 2023; Presidential Ph.D. Fellowship, CUHK, 2019–2023.
- National Scholarship (China), top 0.2% nationwide, 2018; Qiming Star Award (1 of 5 from 7,112 undergraduates), HUST, 2016.
- First Prize, China National Mathematics Olympiad (Province-level Math League), 2012.

Selected Talks

- *LLM-RL Instability and the Training–Inference Mismatch* — SLAI and the Bay Area RL community (2025); *Revisiting Trust Region Policy Optimization in the Era of LLMs* — QingkeAI (2025).
- *Exploration at Scale: Theory, Algorithms & Applications* — MIT, Princeton, INFORMS Annual Meeting, ISMP, RLChina (2024); *Tackling Data Scarcity for Trustworthy Agents* — ETH Zurich (2024).